

value_name, x
ident
integer_literal
index, i, j, n, m

<i>typeconstr</i>	$::=$ unit bool int	
<i>typvar, tv</i>	$::=$ <i>'ident</i>	
<i>typexpr, t</i>	$::=$ <i>typvar</i> <i>typexpr</i> $\rightarrow$ <i>typexpr'</i> <i>typeconstr</i> (<i>typexpr</i>)	S
<i>typscheme, ts</i>	$::=$ (<i>typvar</i> ₁ , .., <i>typvar</i> _n) <i>typexpr</i> generalise (Γ , <i>t</i>)	bind <i>typvar</i> ₁ .. <i>typvar</i> _n in <i>typexpr</i> M
<i>constant, c</i>	$::=$ <i>integer_literal</i> false true () (&&) not	
<i>expr, e</i>	$::=$ <i>value_name</i> <i>constant</i> <i>expr expr'</i> function <i>value_name</i> $\rightarrow$ <i>expr</i> let <i>value_name</i> = <i>expr</i> in <i>expr'</i> (<i>expr</i>) { <i>v/x</i> } <i>e</i>	bind <i>value_name</i> in <i>expr</i> bind <i>value_name</i> in <i>expr'</i> S M
<i>value, v</i>	$::=$ <i>constant</i> function <i>value_name</i> $\rightarrow$ <i>expr</i>	
Γ	$::=$ empty Γ , <i>value_name</i> : <i>typscheme</i>	
<i>formula</i>	$::=$ <i>judgement</i> not (<i>formula</i>) <i>typscheme</i> > <i>t</i> <i>typscheme</i> = <i>typscheme'</i> <i>value_name</i> = <i>value_name'</i>	

<i>terminals</i>	$::=$ $\rightarrow$ function $\vdash$ $\longrightarrow$ $\{$ $\}$
<i>Jtype</i>	$::=$ $value_name : \textit{typscheme} \textbf{in} \Gamma$ $\Gamma \vdash constant : t$ $\Gamma \vdash e : t$
<i>Jop</i>	$::=$ $e \longrightarrow e'$
<i>judgement</i>	$::=$ $Jtype$ Jop
<i>user_syntax</i>	$::=$ $value_name$ $ident$ $integer_literal$ $index$ $typeconstr$ $typvar$ $typexpr$ $typscheme$ $constant$ $expr$ $value$ Γ $formula$ $terminals$

$value_name : \textit{typscheme} \textbf{in} \Gamma$

$$\frac{}{value_name : \textit{typscheme} \textbf{in} \Gamma, value_name : \textit{typscheme}} \quad \text{VTSIN_VN1}$$

$$\frac{\frac{value_name : \textit{typscheme} \textbf{in} \Gamma \quad \textbf{not} (value_name = value_name')}{value_name : \textit{typscheme} \textbf{in} \Gamma, value_name' : \textit{typscheme}'}}{} \quad \text{VTSIN_VN2}$$

$\Gamma \vdash constant : t$

$$\frac{}{\Gamma \vdash integer_literal : \textbf{int}} \quad \text{CONSTANT_INT}$$

$$\frac{}{\Gamma \vdash \textbf{false} : \textbf{bool}} \quad \text{CONSTANT_FALSE}$$

$$\frac{}{\Gamma \vdash \textbf{true} : \textbf{bool}} \quad \text{CONSTANT_TRUE}$$

$$\frac{}{\Gamma \vdash () : \mathbf{unit}} \quad \text{CONSTANT_UNIT}$$

$$\frac{}{\Gamma \vdash (\&\&) : \mathbf{bool} \rightarrow (\mathbf{bool} \rightarrow \mathbf{bool})} \quad \text{CONSTANT_AND}$$

$$\frac{}{\Gamma \vdash \mathbf{not} : \mathbf{bool} \rightarrow \mathbf{bool}} \quad \text{CONSTANT_NOT}$$

$$\boxed{\Gamma \vdash e : t}$$

$$\frac{x : \textit{typscheme} \textbf{ in } \Gamma \quad \textit{typscheme} > t}{\Gamma \vdash x : t} \quad \text{GET_VALUE_NAME}$$

$$\frac{\Gamma \vdash \textit{constant} : t}{\Gamma \vdash \textit{constant} : t} \quad \text{GET_CONSTANT}$$

$$\frac{\Gamma \vdash e : t_1 \rightarrow t_2 \quad \Gamma \vdash e' : t_1}{\Gamma \vdash e e' : t_2} \quad \text{GET_APPLY}$$

$$\frac{\Gamma, x_1 : () t_1 \vdash e : t}{\Gamma \vdash \mathbf{function} x_1 \rightarrow e : t_1 \rightarrow t} \quad \text{GET_LAMBDA}$$

$$\frac{\Gamma \vdash e : t \quad \Gamma, x : \textit{typscheme} \vdash e' : t' \quad \textit{typscheme} = \mathbf{generalise}(\Gamma, t)}{\Gamma \vdash \mathbf{let} x = e \textbf{ in } e' : t'} \quad \text{GET_LET}$$

$$\boxed{e \longrightarrow e'}$$

$$\frac{}{(\mathbf{function} x \rightarrow e) v \longrightarrow \{v/x\}e} \quad \text{JO_RED_APP}$$

$$\frac{}{\mathbf{let} x = v \textbf{ in } e \longrightarrow \{v/x\}e} \quad \text{JO_RED_LET}$$

$$\frac{e \longrightarrow e'}{e e_1 \longrightarrow e' e_1} \quad \text{JO_RED_CONTEXT_APP1}$$

$$\frac{e \longrightarrow e'}{v e \longrightarrow v e'} \quad \text{JO_RED_CONTEXT_APP2}$$

$$\frac{e \longrightarrow e'}{\mathbf{let} x = e \textbf{ in } e_1 \longrightarrow \mathbf{let} x = e' \textbf{ in } e_1} \quad \text{JO_RED_CONTEXT_LET}$$

$$\frac{}{\mathbf{not true} \longrightarrow \mathbf{false}} \quad \text{JO_RED_NOT_1}$$

$$\frac{}{\mathbf{not false} \longrightarrow \mathbf{true}} \quad \text{JO_RED_NOT_2}$$

$$\frac{}{((\&\&) \mathbf{true}) e \longrightarrow e} \quad \text{JO_RED_AND_1}$$

$$\frac{}{((\&\&) \mathbf{false}) e \longrightarrow \mathbf{false}} \quad \text{JO_RED_AND_2}$$

Definition rules: 22 good 0 bad

Definition rule clauses: 36 good 0 bad